

ewe

SE vacuum stability

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June 2024

1 Introduction

1.1 The unperturbed solution

here some historic background on the SE
the equations describing the problem

1.1.1 The self-similar solution

The Euler fluid dynamics equations describe the motion of an inviscid fluid. in
a spherically symmetric density profile $\rho = \kappa r^{-\omega}$, the equations are given by

$$\begin{aligned} 0 &= (\partial_t + u\partial_r) \rho + \frac{\rho}{r^2} \frac{\partial}{\partial r} (r^2 u) \\ 0 &= \rho (\partial_t + u\partial_r) u + \partial_r \left(\frac{\rho c^2}{\gamma} \right) \\ 0 &= (\partial_t + u\partial_r) \left(\frac{c^2 \rho^{1-\gamma}}{\gamma} \right) \end{aligned} \tag{1}$$

We predict that at large t the solution is self-similar, so the variables can be
written as -

$$\begin{aligned} \xi &\equiv \frac{r}{R(t)} \\ u(r, t) &= \dot{R}\xi U(\xi) \quad c(r, t) = \dot{R}\xi C(\xi) \quad \rho(r, t) = BR^\varepsilon G(\xi) \end{aligned} \tag{2}$$

And by inserting these into the equations we get the following ODEs -

$$\begin{aligned} 0 &= [\varepsilon - \xi (\log(G)')] + \xi U (\log(G)') + (\xi U)' + 2U \\ 0 &= [\delta \xi U + \xi (U - 1) (\xi U)'] + \frac{1}{\gamma G} (G \xi^2 C^2)' \\ 0 &= 2\delta - (\gamma - 1) \varepsilon + \xi (U - 1) [\log(\xi^2 C^2) - (\gamma - 1) \log(G)]' \end{aligned} \tag{3}$$

Where $\delta = \frac{R\ddot{R}}{R^2}$.

In order to get a self-similar time-independent solution, we demand that $\delta =$

const. In the shock front we have the boundary conditions from the Rankine-Hugoniot conditions -

$$G(1) = \frac{\gamma+1}{\gamma-1} \quad C(1) = \frac{\sqrt{2\gamma(\gamma-1)}}{\gamma+1} \quad U(1) = \frac{2}{\gamma+1} \quad (4)$$

Using the boundary conditions we can calculate the value of $\varepsilon = -\omega$ in order for the the two sides of the shock to scale correctly.

The PDE can be better understood by rewriting it by the following variables -

$$\begin{aligned} \Delta &= C^2 - (1-U)^2 \\ \Delta_1 &= U(1-U)(1-U-\delta) - C^2(3U + \frac{\varepsilon+2\delta}{\gamma}) \\ \Delta_2 &= C(1-U)(1-U-\delta) - \frac{\gamma-1}{2}CU(2-2U+\delta) - C^2 + \frac{2\delta-(\gamma-1)\varepsilon}{2\gamma} \frac{C^3}{1-U} \end{aligned} \quad (5)$$

And the equation is reduced to following -

$$\frac{dU}{d \log \xi} = \frac{\Delta_1}{\Delta} \quad \frac{dC}{d \log \xi} = \frac{\Delta_2}{\Delta} \quad (6)$$

and G is given by the following -

$$\begin{aligned} G &= \sqrt[\gamma-1+\lambda]{\kappa \cdot C^2 \cdot \xi^{2-3\lambda} (1-U)^{-\lambda}} \\ \lambda &= \frac{2\delta + \omega(\gamma-1)}{3-\omega} \end{aligned} \quad (7)$$

1.1.2 The first type solutions

The first type solutions where discovered by Taylor, Seydov, and Von-Neumann. By taking $\delta = \frac{\omega-3}{2}$ we can solve the equations.

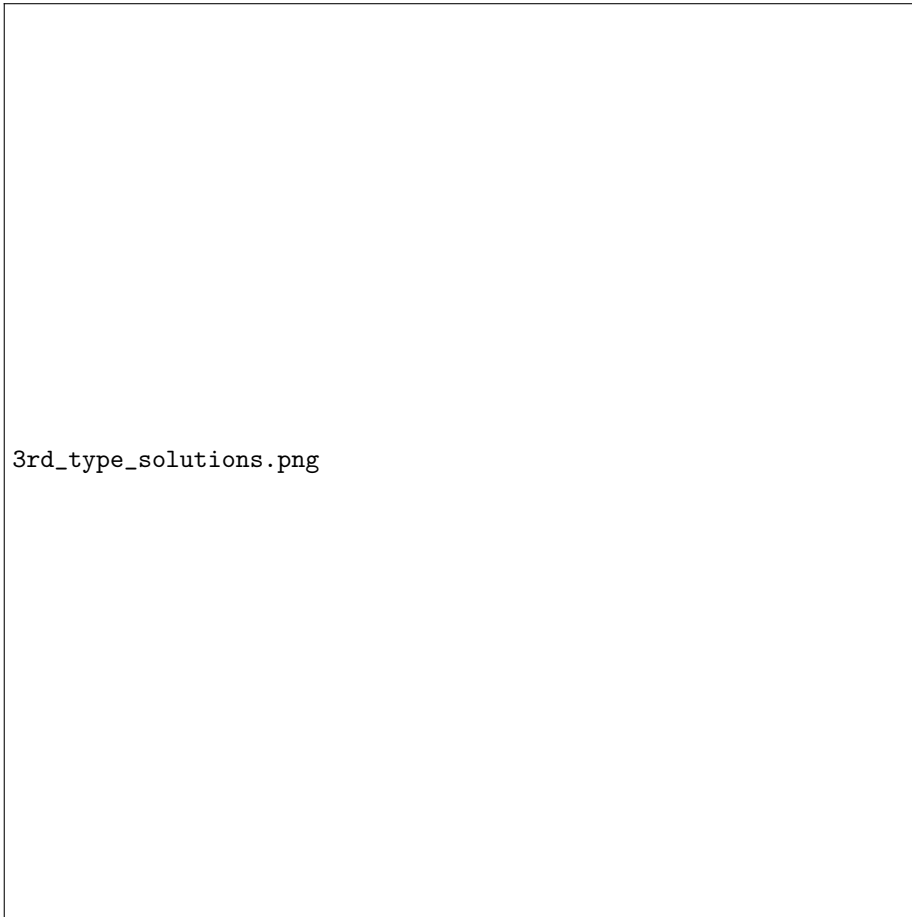
It is important to note that for $\omega \in (2, 3)$ we get a "vacuum" solution, we shall return to this later.

1.1.3 The second type solutions

The second type solutions where...

1.1.4 third type solutions

here some background on the third type solutions, including the work by Gruzinov.



3rd_type_solutions.png

Figure 1: Enter Caption

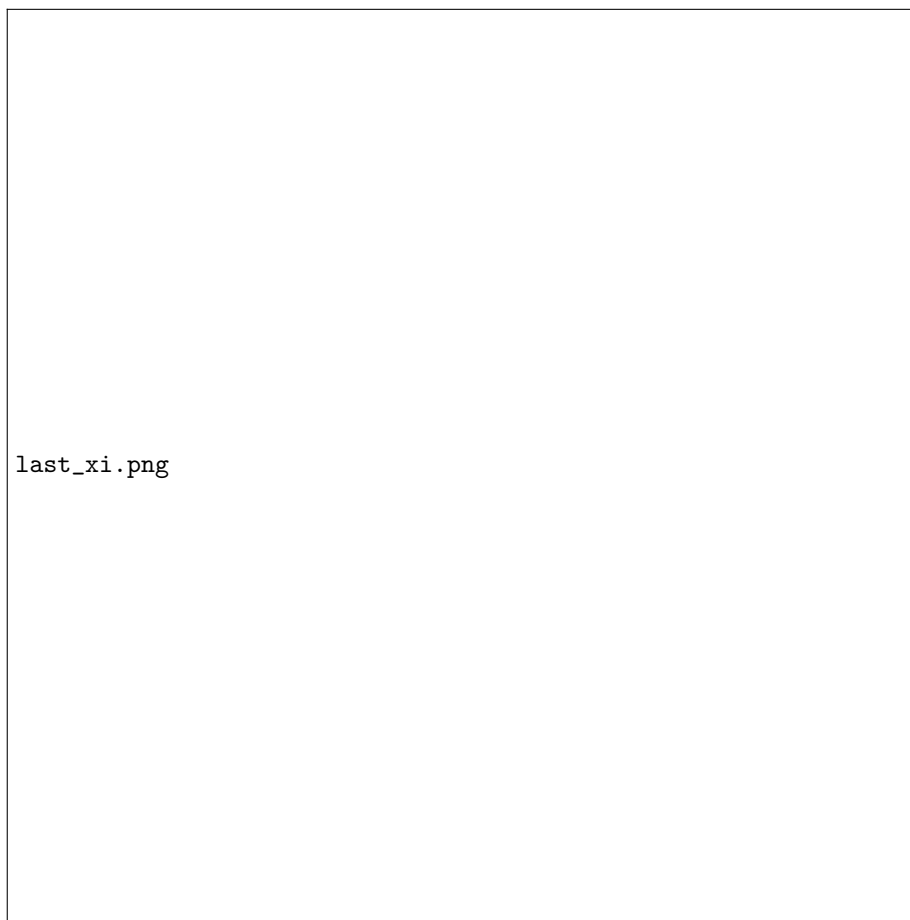


Figure 2: Enter Caption

1.1.5 The vacuum solution

Once the solution reaches $U = 1$, in which cases we have a vacuum in our solution

1.2 The perturbed solution

here some background on the perturbation, including the work by Ream, waxman, j sanz...

1.2.1 the methods used

Let the perturbed variables be given by the following -

$$\begin{aligned}\delta\vec{v} &= \vec{v}(r, \theta, \phi, t) - v_0(r, t) \\ \delta p &= p(r, \theta, \phi, t) - p_0(r, t) \\ \delta\rho &= \rho(r, \theta, \phi, t) - \rho_0(r, t)\end{aligned}\tag{8}$$

And by separation of variables we get the following equations -

$$\begin{aligned}\delta\vec{v}(r, \theta, \phi, t) &= \xi \dot{R} [\delta U_r(\xi) Y_{lm}(\theta, \phi) \hat{r} + \delta U_T(\xi) \nabla_T Y_{lm}(\theta, \phi)] f \\ \delta\rho(r, \theta, \phi, t) &= BR^\varepsilon \delta G(\xi) Y_{lm}(\theta, \phi) f \\ \delta p(r, \theta, \phi, t) &= BR^\varepsilon \delta G(\xi) Y_{lm}(\theta, \phi) f\end{aligned}\tag{9}$$

after inserting these into the equations and taking only first derivatives of the perturbed variables we get the following equations -

$$0 = \frac{\partial \delta\rho}{\partial t} + r^{-2} \frac{\partial}{\partial r} (r^2 v_0 \delta\rho) + \nabla(\rho_0 \delta\vec{v})\tag{10}$$

$$0 = \frac{\partial \delta\vec{v}}{\partial t} + v_0 \frac{\partial \delta\vec{v}}{\partial r} + \delta v_r \frac{\partial v_0}{\partial r} \hat{r} - \rho_0^{-2} \frac{\partial p_0}{\partial r} + \rho_0^{-1} \nabla(\delta p) + r^{-1} v_0 \delta v_t\tag{11}$$

$$0 = \left(\frac{\partial}{\partial t} + v_0 \frac{\partial}{\partial r} \right) \left(\frac{\delta p}{p_0} - \gamma \frac{\delta\rho}{\rho_0} \right) + \delta v_r \left(p_0^{-1} \frac{\partial p_0}{\partial r} - \gamma \rho_0^{-1} \frac{\partial \rho_0}{\partial r} \right)\tag{12}$$

Or in matrix form -

$$MY' = NY\tag{13}$$

where

$$Y = \begin{bmatrix} \delta G \\ \delta U_R \\ \delta U_T \\ \delta P \end{bmatrix} \quad q = \frac{\dot{f}R}{f\dot{R}}\tag{14}$$

$$M = \begin{bmatrix} \xi(U-1) & G\xi & 0 & 0 \\ 0 & (U-1)\xi^2 G & 0 & 1 \\ 0 & 0 & (U-1)\xi^2 G & 0 \\ -\frac{\gamma\xi}{G}(U-1) & 0 & 0 & \frac{\xi}{P}(U-1) \end{bmatrix}\tag{15}$$

$$N = \begin{bmatrix} \omega - q - 3U - \xi U' & -\xi G' - 3G & l(l+1)G & 0 \\ P'G^{-1} & (1-\delta-q-2U-\xi U')G\xi & 0 & 0 \\ 0 & 0 & (1-\delta-q-2U)G\xi & -\xi^{-1} \\ \frac{\gamma q}{G} - \xi\gamma(U-1)\frac{G'}{G^2} & \xi\gamma\frac{G'}{G} & 0 & \xi(U-1)\frac{P'}{P^2} - \frac{q}{P} \end{bmatrix}\tag{16}$$

And the boundary conditions in the shock front are given by the following -

$$\begin{aligned}
\delta G(1) &= -\frac{\gamma+1}{\gamma-1}\omega - G' \\
\delta U_r(1) &= \frac{2}{\gamma+1}q - U' \\
\delta U_T(1) &= -\frac{2}{\gamma+1} \\
\delta P(1) &= \frac{2}{\gamma+1}[2(q+1) - \omega] - P'
\end{aligned} \tag{17}$$

Now, the way to solve this is by integrating the equations from the shock front to the singular point. only for correct values of q we will get a solution that doesn't diverge at the singular point.

This method works very well for the second type solutions, but for the third type solutions we encounter a problem, as the base solution diverges at the singular point.

1.2.2 The perturbed solution for the second type solutions

1.3 Analyzing the singular point in solutions with a vacuum

Near the singular point $C \rightarrow 0$, $U \rightarrow 1$, and $G \rightarrow \infty$.
we will denote U as a function of C -

$$(U - 1) = AC^\theta \tag{18}$$

and by inserting this into the equation and we get the following -

$$\theta(3 \leq \delta \leq 3.26) = \frac{6\gamma - 2\omega}{(\gamma - 1)\omega} \in (1, 2) \tag{19}$$

$$\theta(2 \leq \delta \leq 3) = 2 \tag{20}$$

$$A = \text{const} \dots \tag{21}$$

And the other variables are -

$$G = G_0 \cdot C^{-2} \cdot \xi_{vacuum}^{3-2\omega} \tag{22}$$

$$P = \frac{G_0}{\gamma} \cdot \xi_{vacuum}^{5-2\omega} \tag{23}$$

$$G_0 = \text{const} \dots \tag{24}$$

In order to understand what the perturbations look like near the singular point, we will look at the possible eigenvalues of equation -

$$\det(M - \lambda N) = 0 \tag{25}$$

the matrix is then -

$$\begin{bmatrix} -\xi AC^\theta - \lambda \left[\omega \left(1 - \frac{\xi}{5} \right) + 3(\xi - 1) - q \right] & \gamma P \xi^{-1} \cdot C^{-2} - \lambda \gamma P \frac{2\omega}{5A} \xi^{-1} C^{-2-\theta} & -\lambda \gamma P \cdot l(l+1) \xi^{-2} C^{-2} & 0 \\ -\lambda \left[3 - \frac{6}{5} \omega \right] \xi \cdot C^2 & -A \gamma P C^{\theta-2} - \lambda \gamma P \left(3\xi \left(1 - \frac{\omega}{5} \right) - (1+q) \right) \xi^{-1} C^{-2} & 0 & 1 \\ 0 & 0 & -A \gamma P C^{\theta-2} + \lambda \gamma P (1+q) \xi^{-1} C^{-2} & \lambda \xi^{-1} \\ \frac{A}{P} \xi^3 C^{\theta+2} - \frac{\lambda}{P} \left[q - 2\xi \frac{\omega}{5} \right] \xi^2 C^2 & \lambda \gamma \xi \frac{2\omega}{5A} C^{-\theta} & 0 & -\frac{A}{P} \xi C^\theta + \lambda \frac{q}{P} \end{bmatrix}$$

and the determinant is -

$$\begin{aligned} 0 = & \left[-\xi AC^\theta - \lambda \left[\omega \left(1 - \frac{\xi}{5} \right) + 3(\xi - 1) - q \right] \right] \\ & \times \left[\gamma^2 P C^{-2} \left[\left(C^{-2} \left(\lambda(1+q) \xi^{-1} - AC^\theta - \lambda 3 \left(1 - \frac{\omega}{5} \right) \right) \right. \right. \right. \\ & \left. \left. \left(\lambda q - A \xi C^\theta \right) - \lambda \xi \frac{2\omega}{5A} C^{-\theta} \right) \left(\lambda(1+q) \xi^{-1} - AC^\theta \right) \right] \right] \\ & + \lambda \left[3 - \frac{6}{5} \omega \right] \gamma^2 P \left(\left(1 - \lambda \frac{2\omega}{5A} C^{-\theta} \right) \left(\lambda(1+q) \xi^{-1} - AC^\theta \right) \right. \\ & \left. \left(\lambda q - A \xi C^\theta \right) C^{-2} - \lambda^3 \frac{2\omega}{5A} C^{-\theta} \xi^{-1} l(l+1) \right) - \xi \left[A \xi C^\theta - \lambda \left[q - 2\xi \frac{\omega}{5} \right] \right] \\ & \times \left[\gamma^2 P \cdot \left[\left(AC^\theta + \lambda \left(3\xi \left(1 - \frac{\omega}{5} \right) - (1+q) \right) \xi^{-1} \right) \right. \right. \\ & \left. \left. \left(l(l+1) C^{-2} \right) \lambda^2 \xi^{-2} + \left(AC^{\theta-2} - \lambda(1+q) \xi^{-1} C^{-2} \right) \left(1 - \lambda \frac{2\omega}{5A} C^{-\theta} \right) \right] \right] \end{aligned}$$

1.3.1 finding the eigenvalues

in order to find the eigenvalues we will assume that λ is a function of C , so $\lambda = \lambda_0 C^\alpha$.

Let us divide and conquer,

1. if $\alpha > \theta$ then we get that near the singular point $\alpha = \theta - 2$ and thus a contradiction.
2. if $\alpha = \theta$ then we get the following eigenvalues -

$$\lambda_0 = \frac{A\xi}{q} \quad (26)$$

$$\lambda_0 = \frac{A\xi}{q + 1 - 3\xi \left(1 - \frac{\omega}{5} \right)} \quad (27)$$

$$\lambda_0 = \frac{A\xi}{q + \omega \left(\frac{\xi}{5} - 1 \right) + 3(1 - \xi)} \quad (28)$$

3. if $\theta - 2 < \alpha < \theta$ - then the determinant is given by -

$$0 = \lambda_0^3 C^{3\alpha-4} \left[\omega \left(1 - \frac{\xi}{5} \right) + 3(\xi - 1) - q \right] \left[(1+q) \xi^{-1} - 3 \left(1 - \frac{\omega}{5} \right) \right] q$$

it means that in order to get non zero eigenvalues the values of q have to be one of the following -

$$\begin{aligned} q_1 &= 0 \\ q_2 &= \omega - \xi_{vacuum} \left(\frac{\omega}{5} + 3 \right) - 3 < 0 \\ q_3 &= \xi_{vacuum} \left(3 - \frac{3}{5}\omega \right) - 1 > 0 \end{aligned} \tag{29}$$

which is weird, because it is independent of l .

4. if $\theta - 2 = \alpha$ - then we get the following eigenvalue -

$$\lambda_0 = \frac{\left[\omega \left(1 - \frac{\xi}{5} \right) + 3(\xi - 1) - q \right] \left[(1 + q) \xi^{-1} - 3 \left(1 - \frac{\omega}{5} \right) \right]}{\left[3 - \frac{6}{5}\omega \right] \frac{2\omega}{5A} (1 + q) \xi^{-1}} \tag{30}$$

this seems to be the only option to get an eigenvalue where the perturbations don't diverge more then the base solution as we approach the singular point.

5. if $\theta - 2 > \alpha$ - then we get the following determinant -

$$0 = \lambda_0^4 C^{\alpha-\theta} \left[3 - \frac{6}{5}\omega \right] \frac{2\omega}{5A} (q(1 + q) \xi^{-1})$$

meaning that in order to get any non-zero eigenvalues the values of q have to be $q = -1, 0$

2 The vacuum stability

2.1 the approximations near the singular point